MOHAMED EID

Senior Full-Stack AI Engineer | AI Systems Architect | AI Solutions Architect | Enterprise AI & Digital Transformation

Cairo, Egypt (UAE resident) | Open to remote and relocation (visa sponsorship required) | Available immediately
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SUMMARY

Senior full-stack AI engineer and systems architect with a telecommunications engineering foundation, 15 years across networks, IT, and software, and hands-on software development since 2019. I build AI products end to end: backend, web, mobile, and multi-provider model integration. Sole developer of TAHEEL, a UAE government-transaction platform, and architect and lead engineer of OpenQCore, a model-agnostic AI platform (~210K lines of code across four codebases). Independent researcher on control-theoretic approaches to reliable, stable AI systems, with open-source tools validated on real LLM agents.

SELECTED ACHIEVEMENTS

- Delivered a UAE government-transaction platform (web, Android, iOS, backend, AI and OCR) serving a large customer base in production.
- Architected OpenQCore, a model-agnostic AI platform: 279 REST endpoints, routing across OpenAI, Anthropic, and Gemini with automatic fallback.
- Cut time-to-first-token by 62% (7.9s to 3.0s) and built graceful model downgrade when user credits run out.
- Two preprints on control-theoretic design and delay-induced instability of AI systems; the second is under peer review at a machine-learning journal.
- Released two open-source verification tools: a stability analyzer for multi-agent loops that predicted every confident outcome correctly on 54 real gpt-4o-mini runs, and a distributed authorization harness verified by mutation testing and an exhaustive model check (2.5M states).

CORE SKILLS

Languages: Python (primary), TypeScript/JavaScript, C and C++; also Java

Hardware and embedded: embedded systems and microcontrollers, board-level repair and soldering, computer and network hardware, telecom equipment installation and troubleshooting

Backend: FastAPI, async Python, REST API design, PostgreSQL, SQLAlchemy 2.0, Alembic, Redis, SQLite

Frontend and mobile: Next.js, React, React Native (Expo), Android and iOS

AI: multi-provider LLM integration (OpenAI, Anthropic, Gemini), multi-agent systems, RAG, embeddings (sentence-transformers, scikit-learn), speech-to-text and text-to-speech, OCR and document intelligence, fine-tuning data pipelines

Management: change management (Lewin, McKinsey 7-S), operations and inventory planning (EOQ, reorder point, JIT, ABC), strategic analysis (SWOT), Six Sigma and root-cause problem solving, crisis management, stakeholder and budget management

Reliability and verification: stability analysis of feedback loops, model checking, mutation testing, property-based and Monte Carlo testing, GitHub Actions CI

Infrastructure and security: Docker, OpenTelemetry, Prometheus, Grafana, Loki, Tempo, JWT, OAuth2, TOTP MFA, Argon2, Ed25519

Also worked with: Kafka, PyTorch, TensorFlow, LangChain, Kubernetes, AWS, Azure, GCP

EXPERIENCE

Lead AI Systems Architect & Engineer | OpenQCore, Tirana, Albania — Dec 2025 – Present

- Architected and built a model-agnostic AI platform (final pre-launch stage): ~93K lines of Python and 279 REST endpoints, with routing across OpenAI, Anthropic, and Gemini and automatic fallback to a default model when a provider fails.
- Cut time-to-first-token from 7.9s to 3.0s (~62%), measured before and after on the running system: moved heavy pre-model work behind the first streamed event, moved message persistence to the background, and moved the database from US East to Frankfurt next to the API server.
- Built RAG and memory services, voice (STT/TTS), image generation, and document intelligence (OCR, PDF/Office parsing, QR/barcode), with Arabic text support.
- Implemented usage-based billing: wallet balances, tiered limits, token-based pricing, and graceful downgrade to a lower-cost model when credits run out.
- Secured and operated the platform: JWT, OAuth2, TOTP MFA, Argon2; Docker deployment with OpenTelemetry, Prometheus, Loki, Tempo, and Alertmanager.
- Built three clients on the same API: a control dashboard (Next.js 16, React 19, 9 languages with RTL), ChatQXT web chat, and QChat, a React Native (Expo) mobile app.
- Built an automated fine-tuning data pipeline with approval gates, dataset builder, model registry, and fine-tuning jobs on external model APIs.

Technical Consultant (on call) | TAHEEL — 2026 – Present

Sole Full-Stack Developer | TAHEEL, UAE government-transaction platform 2024 – 2026

- Designed and built the entire platform: web (Next.js, Firebase), Android and iOS apps, backend, and AI layer, serving a large customer base.
- Moved government-service applications online: customers apply 24/7, and each request is routed automatically to the responsible ministry specialist with customer data and documents attached, so turnaround equals government processing time only, with no branch visits or queues.
- Built request tracking, push notifications (e.g., missing-document alerts), an AI assistant that creates and tracks requests and answers customers, and OCR-based document intake with secure storage.
- Continue to advise the company on an on-call basis.

Team Manager, Technology & Operations (part-time) | IV-CORE LLC, Ajman, UAE Feb 2023 – Dec 2024

- Led process-improvement initiatives, increasing operational efficiency by 20%; managed budgets, resources, and stakeholders.

Earlier roles 2011 – 2023

- Independent Full-Stack Developer (Freelance), 2019–2023: built web and mobile applications end to end for individual clients (frontend, backend, and database design), including Cinema Club, a video-streaming app, and Zed, a platform with a companion app. Began working with AI as the field was emerging.
- Assistant Operations Manager / Technology Lead, Royal Towers and EASY GO (Tbilisi, Georgia, 2018–2023): owned technology infrastructure, data security, and system upgrades; improved system efficiency by 15% and cut operating costs by 10%.
- Telecommunication Site Engineer, Ooredoo (Kuwait City, 2015–2018): installed, maintained, and troubleshot telecom networks; improved network performance by 10% and reduced downtime by 5%.
- LC Engineer, Compu Sciences (Cairo, 2011–2015): IT and network support, system reliability and data protection, including a year of hands-on hardware work: installation, diagnosis and repair of computer and network hardware, board-level repair and soldering, and embedded and microcontroller work.

OPEN-SOURCE PROJECTS & PORTFOLIO

- **loopstab: stability analysis for multi-agent AI loops** (github.com/Mo-eid200/agent-loop-stability): predicts from recorded traces or measured latencies whether an agent loop converges, oscillates or diverges, and recommends safe gains. On 54 real gpt-4o-mini runs it recovered the loop delay in all 54 without being told, with 0 wrong confident verdicts; 12-test verification suite, 11 planted bugs all caught, continuous integration.
- **Final-boundary authorization harness** (github.com/Mo-eid200/final-boundary-harness): reference design ensuring a revoked AI-agent authorization can never execute; distributed services with durable state, challenge-response leases and proof of possession, tested under crashes and network partitions, with mutation testing and an exhaustive model check (all safety properties hold in 2.5M states). Built in public collaboration with an AI-evaluation researcher whose review findings shaped v2.
- **OpenQCore platform overview** (full architecture, design decisions, research): github.com/Open-QCore/openqcore-platform-overview. Source code is private; code walkthrough available on request.

RESEARCH

Independent researcher, OpenQCore | *ORCID 0009-0008-8404-978X*

- Eid, M. (2026). *Delay-Induced Instability in Multi-Agent LLM Orchestration: Exact Stability Boundaries and a Delay-Aware Extension of UCCF.* Preprint, Zenodo. DOI: 10.5281/zenodo.23697430. **Under peer review at a machine-learning journal.** Exact mean-square and almost-sure stability boundaries for delayed multi-agent loops, a proof that the worst-case delay margin is tight for heterogeneous delays, and a delay-aware controller. The version under review adds held-out plant variants and experiments with real gpt-4o-mini agents in the loop, with 33 pre-registered claims and all outcomes reported.
- Eid, M. (2026). *Artificial Intelligence Needs Control Theory: Towards Stable, Adaptive, and Autonomous AI Systems.* Preprint, Zenodo. DOI: 10.5281/zenodo.23378509. Formal closed-loop model with stability conditions, an evaluation framework, and a reproducible Monte-Carlo study (500 seeds per variant).

EDUCATION

- **MSc, Information Technology**, University of the People, USA (in progress)
- **MBA**, University of the People, USA (2019 – 2023). Capstone: a change-management initiative for Nokia, diagnosing its decline (bureaucracy, leadership, management–employee relations) and designing the change process, governance, monitoring and quality assurance using Lewin's three-phase and McKinsey 7-S models.
- **BEng, Electronic & Electrical Telecommunication Engineering**, Cairo University, Egypt (2005 – 2010; graduation project with Distinction; coursework in embedded systems and microcontrollers)
- **BIS, Information Systems**, Pharaohs Academy, Egypt (2006 – 2010; project with Distinction)

Certifications: Cisco CCNA, CCNP **Languages:** Arabic (native), English (fluent)